

LSWT Hamiltonian: Code Convention Derivation

For internal verification

1 Basis transformation

The code defines

$$\mathbf{C} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 & 0 \\ i & -i & 0 \\ 0 & 0 & \sqrt{2} \end{pmatrix}, \quad \mathbf{C}^\dagger \mathbf{C} = \mathbf{I}, \quad \mathbf{C}^T \mathbf{C} = \mathbf{S} \equiv \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}. \quad (1)$$

An important identity is

$$\mathbf{C}^* = \mathbf{C} \mathbf{S}, \quad \mathbf{C}^T = \mathbf{S} \mathbf{C}^\dagger. \quad (2)$$

The rotated exchange matrix in the lab frame is

$$\mathbf{R} \mathbf{J}_{\sigma\rho} = \mathbf{R}_\sigma^T \mathbf{J}_{\sigma\rho} \mathbf{R}_\rho, \quad (3)$$

where $\mathbf{R}_\sigma$ is the rotation from the lab frame to the local spin frame at sublattice σ . The code computes

$$\boxed{\text{hop_t} = \mathbf{C}^\dagger \mathbf{R} \mathbf{J} \mathbf{C}}. \quad (4)$$

2 Explicit elements of hop_t

Direct calculation of $\text{hop_t} = \mathbf{C}^\dagger \mathbf{R} \mathbf{J} \mathbf{C}$ gives:

$$\text{hop_t}_{00} = \frac{1}{2} (\mathbf{R} \mathbf{J}^{xx} + \mathbf{R} \mathbf{J}^{yy} - i(\mathbf{R} \mathbf{J}^{xy} - \mathbf{R} \mathbf{J}^{yx})), \quad (5)$$

$$\text{hop_t}_{01} = \frac{1}{2} (\mathbf{R} \mathbf{J}^{xx} - \mathbf{R} \mathbf{J}^{yy} - i(\mathbf{R} \mathbf{J}^{xy} + \mathbf{R} \mathbf{J}^{yx})), \quad (6)$$

$$\text{hop_t}_{10} = \frac{1}{2} (\mathbf{R} \mathbf{J}^{xx} - \mathbf{R} \mathbf{J}^{yy} + i(\mathbf{R} \mathbf{J}^{xy} + \mathbf{R} \mathbf{J}^{yx})), \quad (7)$$

$$\text{hop_t}_{11} = \frac{1}{2} (\mathbf{R} \mathbf{J}^{xx} + \mathbf{R} \mathbf{J}^{yy} + i(\mathbf{R} \mathbf{J}^{xy} - \mathbf{R} \mathbf{J}^{yx})), \quad (8)$$

$$\text{hop_t}_{02} = \frac{1}{\sqrt{2}} (\mathbf{R} \mathbf{J}^{xz} - i \mathbf{R} \mathbf{J}^{yz}), \quad \text{hop_t}_{12} = \frac{1}{\sqrt{2}} (\mathbf{R} \mathbf{J}^{xz} + i \mathbf{R} \mathbf{J}^{yz}), \quad (9)$$

$$\text{hop_t}_{20} = \frac{1}{\sqrt{2}} (\mathbf{R} \mathbf{J}^{zx} + i \mathbf{R} \mathbf{J}^{zy}), \quad \text{hop_t}_{21} = \frac{1}{\sqrt{2}} (\mathbf{R} \mathbf{J}^{zx} - i \mathbf{R} \mathbf{J}^{zy}), \quad (10)$$

$$\text{hop_t}_{22} = \mathbf{R} \mathbf{J}^{zz}. \quad (11)$$

Hermiticity. When $\mathbf{R} \mathbf{J}$ is a real matrix, hop_t is Hermitian: $\text{hop_t}_{01} = \text{hop_t}_{10}^*$, $\text{hop_t}_{02} = \text{hop_t}_{20}^*$, etc.

3 The code basis: $q = \mathbf{C}^\dagger \bar{S}$

The spin operator in the local frame is $\bar{S} = (\bar{S}^x, \bar{S}^y, \bar{S}^z)^T$. The code implicitly works in the basis

$$q \equiv \mathbf{C}^\dagger \bar{S} = \begin{pmatrix} q^0 \\ q^1 \\ q^2 \end{pmatrix} = \begin{pmatrix} (\bar{S}^x - i\bar{S}^y)/\sqrt{2} \\ (\bar{S}^x + i\bar{S}^y)/\sqrt{2} \\ \bar{S}^z \end{pmatrix} = \begin{pmatrix} \tilde{S}^-/\sqrt{2} \\ \tilde{S}^+/\sqrt{2} \\ \tilde{S}^0 \end{pmatrix}. \quad (12)$$

Here $\tilde{S}^\pm = \bar{S}^x \pm i\bar{S}^y$ are the standard raising/lowering operators. Note the ordering:

$$q^0 = \frac{\tilde{S}^-}{\sqrt{2}} \propto a^\dagger \quad (\text{creation}), \quad q^1 = \frac{\tilde{S}^+}{\sqrt{2}} \propto a \quad (\text{annihilation}). \quad (13)$$

4 Bilinear form in the q -basis

The spin–spin interaction in the local frame is the bilinear form:

$$H_{\sigma\rho} = \bar{S}_\sigma^T \text{RJ}_{\sigma\rho} \bar{S}_\rho. \quad (14)$$

Since $\bar{S} = \mathbf{C} q$ (inverting eq. (??) using $\mathbf{C}^\dagger \mathbf{C} = \mathbf{I}$):

$$H_{\sigma\rho} = (\mathbf{C} q_\sigma)^T \text{RJ} (\mathbf{C} q_\rho) = q_\sigma^T \underbrace{\mathbf{C}^T \text{RJ} \mathbf{C}}_M q_\rho. \quad (15)$$

Using $\mathbf{C}^T = \mathbf{S} \mathbf{C}^\dagger$ (eq. (??)):

$$M = \mathbf{C}^T \text{RJ} \mathbf{C} = \mathbf{S} \underbrace{\mathbf{C}^\dagger \text{RJ} \mathbf{C}}_{\text{hop.t.}} = \mathbf{S} \cdot \text{hop.t.} \quad (16)$$

So $M_{ab} = \text{hop.t}_{S(a),b}$, where S swaps indices $0 \leftrightarrow 1$:

$$M_{0b} = \text{hop.t}_{1b}, \quad M_{1b} = \text{hop.t}_{0b}, \quad M_{2b} = \text{hop.t}_{2b}. \quad (17)$$

The bilinear is then $H = \sum_{a,b} q_\sigma^a M_{ab} q_\rho^b$, and the coefficient of $q_\sigma^a q_\rho^b$ is M_{ab} .

5 Holstein–Primakoff mapping

From eq. (??) and the standard HP transformation ($\tilde{S}^- \approx \sqrt{2S} a^\dagger$, $\tilde{S}^+ \approx \sqrt{2S} a$):

$$q^0 = \frac{\tilde{S}^-}{\sqrt{2}} \approx \sqrt{S} a^\dagger, \quad q^1 = \frac{\tilde{S}^+}{\sqrt{2}} \approx \sqrt{S} a, \quad q^2 = \tilde{S}^0 \rightarrow S \text{ (classical)}. \quad (18)$$

Therefore each bilinear $q^a q^b$ maps to bosonic operators as:

q -pair	Boson operator	Coupling M_{ab}	= <code>hop.t</code> element
$q^0 q^0$	$S a^\dagger a^\dagger$ (pair creation)	$M_{00} = \text{hop.t}_{10}$	<code>hop.t</code> [1,0]
$q^1 q^1$	$S a a$ (pair annihilation)	$M_{11} = \text{hop.t}_{01}$	<code>hop.t</code> [0,1]
$q^0 q^1$	$S a^\dagger a$ (normal hopping)	$M_{01} = \text{hop.t}_{11}$	<code>hop.t</code> [1,1]
$q^1 q^0$	$S a a^\dagger$ (normal, reordered)	$M_{10} = \text{hop.t}_{00}$	<code>hop.t</code> [0,0]
$q^0 q^2$	$\sqrt{S} S a^\dagger$ (linear, site σ)	$M_{02} = \text{hop.t}_{12}$	<code>hop.t</code> [1,2]
$q^2 q^0$	$S \sqrt{S} a^\dagger$ (linear, site ρ)	$M_{20} = \text{hop.t}_{20}$	<code>hop.t</code> [2,0]

6 BdG Hamiltonian matrix

The BdG structure (PDF eq. 40, 45):

$$\hat{H}_{\mathbf{k}} = \begin{pmatrix} A_{\mathbf{k}} & B_{\mathbf{k}} \\ B_{\mathbf{k}}^\dagger & A_{-\mathbf{k}}^* \end{pmatrix}, \quad \Psi_{\mathbf{k}}^\dagger = (a_{\mathbf{k},1}^\dagger, \dots, a_{\mathbf{k},N_s}^\dagger, a_{-\mathbf{k},1}, \dots, a_{-\mathbf{k},N_s}). \quad (19)$$

- Upper-right $[\sigma, N_s + \rho] = B_{\mathbf{k}}$: couples $a_{\mathbf{k},\sigma}^\dagger a_{-\mathbf{k},\rho}^\dagger = \text{pair creation} \Rightarrow \text{coefficient } M_{00} = \text{hop_t}_{10}$.
- Lower-left $[N_s + \sigma, \rho] = B_{\mathbf{k}}^\dagger$: Hermitian conjugate of $B_{\mathbf{k}}$.

Code assignment. The code defines $\text{tpp} = \sqrt{S_\sigma S_\rho} \text{hop_t}_{10}$ and assigns:

$$H[\sigma, N_s + \rho] += \text{tpp} \cdot e^{-i\mathbf{k} \cdot \boldsymbol{\delta}} \quad (\text{B block, upper-right}), \quad (20)$$

$$H[N_s + \rho, \sigma] += (\text{tpp} \cdot e^{-i\mathbf{k} \cdot \boldsymbol{\delta}})^* \quad (\text{B}^\dagger \text{ block, lower-left}). \quad (21)$$

This is **correct**: the pair creation coupling $M_{00} = \text{hop_t}_{10}$ appears in the B block.

Normal hopping. The code defines $\text{tpm} = \sqrt{S_\sigma S_\rho} \text{hop_t}_{11}$:

$$H[\sigma, \rho] += \text{tpm} \cdot e^{-i\mathbf{k} \cdot \boldsymbol{\delta}} \quad (\text{A block}), \quad (22)$$

$$H[\rho, \sigma] += (\text{tpm} \cdot e^{-i\mathbf{k} \cdot \boldsymbol{\delta}})^* \quad (\text{Hermitian conjugate}). \quad (23)$$

This is **correct**: the normal hopping coupling $M_{01} = \text{hop_t}_{11}$ appears in the A block.

7 Linear terms

The linear term is the coefficient of $a_\sigma^\dagger$ in the Hamiltonian, which must vanish at classical equilibrium.

From exchange couplings. For site σ from the $q_\sigma^0 q_\rho^2$ coupling (with $q_\rho^2 \rightarrow S_\rho$ classically):

$$\text{coupling } M_{02} = \text{hop_t}_{12}, \quad q_\sigma^0 \approx \sqrt{S_\sigma} a_\sigma^\dagger. \quad (24)$$

For site ρ from $q_\sigma^2 q_\rho^0$:

$$\text{coupling } M_{20} = \text{hop_t}_{20}, \quad q_\rho^0 \approx \sqrt{S_\rho} a_\rho^\dagger. \quad (25)$$

Code assignment.

$$f_\sigma += \sqrt{2} S_\rho \text{hop_t}_{12}^{(\sigma\rho)}, \quad (26)$$

$$f_\rho += \sqrt{2} S_\sigma \text{hop_t}_{20}^{(\sigma\rho)}. \quad (27)$$

The hop_t indices $[1,2]$ and $[2,0]$ match M_{02} and M_{20} respectively. The $\sqrt{2}$ prefactor arises from the relation $\text{RJ}^{xz} + i \text{RJ}^{yz} = \sqrt{2} \text{hop_t}_{12}$ (eq. (8)), converting between the Cartesian and q -basis normalizations. At classical equilibrium, $f_\sigma = 0$ for all sublattices.

8 Comparison with the PDF note and legacy code

PDF note (eq. 28). The PDF defines $\tilde{J}^{\alpha\beta}$ with $\alpha, \beta \in \{+, -, 0\}$ via $(C^\dagger \text{RJ} C)_{\alpha\beta}$ (eq. 27), with indices ordered as $(+, -, 0)$ in the matrix display. However, the q -basis (eq. (??)) has ordering $(-, +, 0)$ for its indices $(0, 1, 2)$. Comparing with the explicit formulas in eq. 28:

$$\text{hop_t}_{00} = \tilde{J}^{-+}, \quad \text{hop_t}_{01} = \tilde{J}^{--}, \quad \text{hop_t}_{10} = \tilde{J}^{++}, \quad \text{hop_t}_{11} = \tilde{J}^{+-}, \quad \text{hop_t}_{22} = \tilde{J}^{00}. \quad (28)$$

The PDF bilinear (eq. 23) uses $\tilde{J}^{\bar{\alpha},\beta}$ with $\bar{+} = -, \bar{-} = +$. This $\bar{\alpha}$ convention compensates for the $(+, -, 0)$ labeling vs the q -basis $(-, +, 0)$ ordering:

$$\text{coupling for } q^a q^b = \tilde{J}^{\bar{\alpha}(a), \hat{\beta}(b)} = M_{ab} = \text{hop_t}_{S(a),b}, \quad (29)$$

where $\hat{\alpha}(0) = -, \hat{\alpha}(1) = +$, and $\bar{-} = +$, so $\bar{\alpha}(0) = +$ selects row 0 of $\tilde{J}$, which is row 1 of hop_t (via the S-swap). All approaches give the same result.

Legacy code. The legacy code uses the same $\text{tpp} = \text{hop_t}[1,0]$ but places it in the **lower-left** block ($B^\dagger$ position) with its conjugate in the upper-right:

$$\text{Legacy: } H[N_s + \sigma, \rho] += \text{tpp} \cdot e^{-i\mathbf{k} \cdot \boldsymbol{\delta}} \quad (\text{lower-left}), \quad (30)$$

$$H[\rho, N_s + \sigma] += \text{tpp}^* \cdot e^{+i\mathbf{k} \cdot \boldsymbol{\delta}} \quad (\text{upper-right}). \quad (31)$$

This swaps B and $B^\dagger$: the upper-right (pair creation) block receives $\text{hop_t}_{10}^* = \text{hop_t}_{01}$ instead of hop_t_{10} . For the stripe phase where RJ is real symmetric and $\text{hop_t}_{10} = \text{hop_t}_{01}^*$ is real, $B = B^\dagger$ and the swap has no effect. For non-collinear phases (e.g., NBCP) where $\text{hop_t}_{10} \neq \text{hop_t}_{10}^*$, the legacy code produces incorrect results.

9 Summary

Physical quantity	Coupling	hop_t index	Code variable
Pair creation (B block, $a^\dagger a^\dagger$)	M_{00}	[1,0]	tpp
Normal hopping (A block, $a^\dagger a$)	M_{01}	[1,1]	tpm
Diagonal ($S^z S^z$)	M_{22}	[2,2]	t00
Linear (site σ , $a_\sigma^\dagger$)	M_{02}	[1,2]	—
Linear (site ρ , $a_\rho^\dagger$)	M_{20}	[2,0]	—

All code indices are correct. The relationship $M = \mathbf{S} \cdot \text{hop_t}$ (eq. (??)) provides the systematic mapping between the physical coupling matrix and the code's hop_t elements.